PEIYI LIU

peiyiliu@std.uestc.edu.cn · liupeiyi0101.github.io

RESEARCH INTERESTS

Mathematical Theory of Deep Learning · Generative Models · Numerical Analysis · Convergence Theory · Differential Equations

EDUCATION

University of Electronic Science and Technology of China (UESTC) **Expected Jun 2027**
M.S. in Mathematics | Sep 2024 – Present

Guangxi University **Jun 2023**
B.S. in Applied Mathematics | Sep 2019 – Jun 2023

PUBLICATION AND PREPRINT

Peiyi Liu, Zhaoqiang Liu, and Yiqi Gu. “Operator splitting based diffusion samplers and improved convergence analysis.”
arXiv:2601.17375 (2026). https://arxiv.org/abs/2601.17375
Under review at IEEE Transactions on Information Theory.

RESEARCH EXPERIENCE

Mathematical Analysis of Diffusion-Model Samplers **2024 – Present**
School of Mathematical Sciences, UESTC | Graduate research

- Developed an operator-splitting approach to probability flow ODE sampling, separating linear drift from nonlinear score-driven dynamics and constructing second-order and higher-order schemes.
- Analyzed a Strang-splitting-based sampler using numerical analysis and probability tools; derived non-asymptotic total variation convergence bounds with improved data-dimension dependence.
- Quantified the contributions of numerical discretization, score estimation error, and score-Jacobian approximation error to sampling accuracy.
- Conducted numerical experiments on two-dimensional synthetic data to examine the predicted second-order convergence rate.

MUV-Net and Multimodal MUV-Net | Medical Ultrasound Imaging Research

- Independently designed the MUV-Net architecture for patient-level lymph node metastasis prediction from whole-lesion B-mode ultrasound videos, using segmentation-guided inputs and attention-based multiple-instance learning.
- Designed a multimodal extension integrating B-mode ultrasound videos, color Doppler images, and strain elastography through dedicated ResNet-18 encoders and patient-level feature fusion.
- Implemented the complete network architectures and training pipelines in code, and independently conducted model training for both the original and multimodal MUV-Net systems.

TEACHING EXPERIENCE

Teaching Assistant, Linear Algebra **Sep 2025 – Jan 2026**
University of Glasgow – UESTC Collaborative Educational Program

- Held weekly student consultation and problem-solving sessions; led focused review and question-and-answer sessions ahead of midterm and final examinations.

Teaching Assistant, Calculus **Feb 2025 – Jun 2025**
University of Glasgow – UESTC Collaborative Educational Program

- Provided weekly academic support and addressed students’ questions; delivered examination-period review and question-and-answer sessions.

HONORS AND AWARDS

Academic Scholarship, UESTC **2026**

TECHNICAL SKILLS

Programming and Scientific Computing: Python, MATLAB | Mathematical Typesetting: LaTeX